

# HARSH SINGH

AI / ML & GENERATIVE AI ENGINEER

Delhi-NCR, India | [singhharsh8118@gmail.com](mailto:singhharsh8118@gmail.com) | +91 844 769 5840 | [LinkedIn](#) | [GitHub](#)

Motivated AI/ML Enthusiast with deep hands-on experience in production-grade machine learning, deep learning, and Generative AI. Expert in building end-to-end data pipelines, statistical modeling, and deploying scalable ML solutions using PyTorch, HuggingFace Transformers, and Azure ML. Strong track record of architecting RAG/GraphRAG pipelines, optimizing semantic embeddings, and fine-tuning LLMs (OpenAI, LLaMA, Mistral). Adept at translating complex analytical results into clear business insights for cross-functional stakeholders.

## EDUCATION

**[DTU] Delhi Technological University**  
(Formerly Delhi College of Engineering)  
B.Tech, Electrical Engineering | 2022 – 2026

## TECHNICAL SKILLS

**Programming & Libraries:** Python, SQL, HuggingFace Transformers

**ML/DL:** scikit-learn, TensorFlow, Keras, PyTorch

**Generative AI:** RAG, LLMs, LangChain, LangGraph, FAISS, Tavily, OpenAI Whisper, LLM Fine-tuning

**Cloud & MLOps:** Azure, Docker, Kubernetes, CI/CD

**Visualization:** Tableau, Power BI, Matplotlib, Seaborn

**Data:** DBMS, EDA, Data Cleaning & Validation

## PROFESSIONAL EXPERIENCE

### Industrial Trainee — Data Analytics | NTPC, Dadri

- Analyzed operational and performance data from regenerative feed heating systems as part of a team project focused on predicting electricity consumption, applying structured data cleaning and exploratory analysis to support the group's forecasting effort.
- Analyzed real-time SCADA monitoring systems end-to-end to understand sensor data pipelines, gaining exposure to data quality, reliability, and governance considerations for mission-critical, predictive-maintenance systems.
- Collaborated with a cross-functional trainee team to prepare, validate, and document datasets used in the group's electricity consumption prediction model.

## PROJECTS

### DocuChat — RAG Q&A Chatbot — [Live App](#)

*Retrieval-augmented document Q&A system*

**Tools:** Python, LangChain, FAISS, OpenAI, Streamlit, Docker, PyTorch, HuggingFace Transformers

- Engineered an end-to-end RAG chatbot using LangChain's ConversationalRetrievalChain, powered by GPT-3.5-turbo with a FAISS vector store and OpenAI embeddings (text-embedding-ada-002), enabling users to query PDF/TXT documents and receive answers cited by filename and page number.
- Implemented contextual compression (LLMChainExtractor) to prune retrieved chunks down to the relevant spans, and MMR re-ranking to reduce redundancy across the top-K retrieved documents before passing context to the LLM.
- Tuned retrieval granularity using a RecursiveCharacterTextSplitter (1000-character chunks, 150-character overlap) to balance context completeness against retrieval precision.
- Added conversational memory (ConversationBufferWindowMemory, last 6 turns) so the chatbot could handle multi-turn follow-up questions rather than treating each query in isolation.
- Containerized the app with Docker and deployed a dark-themed Streamlit UI, with documented deployment paths for Streamlit Community Cloud, Railway, and Render.

### AI Research Agent — [Live App](#)

*Stateful, human-in-the-loop research agent*

**Tools:** Python, LangGraph, LangChain, Tavily, Streamlit

- Built a stateful research agent as an 8-node LangGraph StateGraph (planner → review\_plan → researcher → writer → critic → refine → review\_report → finalize) that plans, searches, and synthesizes a cited report on a given topic.
- Implemented two human-in-the-loop checkpoints using LangGraph's interrupt()/Command(resume=...), letting a user approve, edit, or reject both the research plan and the final report before the agent proceeds.

- Built a self-correction loop where a critic node scores each report draft and triggers additional research rounds (bounded by a configurable revision cap, MAX\_REVISIONS) when topic coverage falls below a configurable quality threshold (QUALITY\_THRESHOLD).
- Grounded every claim in real-time web evidence via the Tavily search API, with inline citations and a verified Sources list assembled directly from agent state rather than generated by the LLM.
- Added exponential-backoff retry logic (tenacity) for search/LLM calls and a fully offline mock mode, backed by a pytest suite that runs the entire graph end-to-end with zero API calls.

## VulgarVeto — [Live App](#)

*Audio profanity detection & censorship tool*

Tools: Python, OpenAI Whisper, pydub, gTTS, Streamlit

- Built VulgarVeto, a tool that transcribes uploaded audio using OpenAI's Whisper model, flags words against a configurable 18-term profanity list, and regenerates a censored version of the audio using text-to-speech (gTTS).
- Implemented chunked audio processing with pydub, splitting audio into user-adjustable 5–30 second segments (default 15s) so longer files could be transcribed and reassembled without memory or performance issues.
- Built a live "profanity meter" that computes and displays the percentage of flagged words per upload, with 4 selectable censorship styles (Beep, Random — drawing from 7 alternate tokens, Dolphin, Symbols).
- Designed an interactive Streamlit interface with real-time progress indicators and a side-by-side view of the original transcription versus the censored output for user transparency.

## LEADERSHIP & COLLABORATION

---

- **Public Relations, TEDxDTU (“AI Boom or Bane”):** Managed outreach and stakeholder coordination for a 500+ attendee event, building working relationships across multiple teams.
- **Logistics Head, ENGIFEST, DTU:** Directed logistics for DTU's annual festival, coordinating a 30+ member volunteer team and multiple stakeholders under time pressure.
- **Volunteer, RobinHood Army (Delhi-NCR):** Taught and mentored underprivileged children, contributing 40+ hours of community service; developed strong communication skills for simplifying concepts to varied audiences.

## ACCOMPLISHMENTS

---

- Cleared SSB screening and advanced through the multi-day officer selection assessment for the Indian Armed Forces.
- Secured AIR 11,085 out of 2.1M+ candidates in the 2022 JEE national engineering entrance examination.
- 1st Place — Annual Inter-College Sports Meet, IIT Mandi (2023).